

Analytical Report:
Predicting Voting Outcome for Legalizing Gambling

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Executive Summary

In this report, I use the data mining tool KNIME to build and compare predictive models for a dataset of voting data, containing demographic data of U.S. counties and voting outcomes, to predict voting outcomes for legalizing gambling. The five predictive models that were chosen are Decision Tree, Random Forest, Boosted Tree, Artificial Neural Network, and Logistic Regression. The final objectives are to compare model performances, and to make suggestions for how the models can be utilized for predicting voting outcomes for legalizing gambling.

This report begins with a statement about the business objectives, namely to select a model for predicting voting outcomes for legalizing gambling in order to identify U.S. counties where policies to legalize gambling are more or less likely to pass. The dataset is relatively small, with only 1,287 data points, with each data point representing a single U.S. county. The report moves on to explain the data preprocessing step, where feature engineering, data balancing, and other data preprocessing techniques are applied to prepare the data for modeling.

I then use KNIME to input the preprocessed data into the five chosen models, and use the models to generate their own sets of predictions. I use Scorer and ROC Curve nodes to compare the models' performances based on metrics such as accuracy, sensitivity, specificity, and AUC value, and conclude that the Boosted Tree model performs the best in predicting voting outcome. I conclude the report with suggestions to test the models on larger datasets of voter data if possible, to apply the Boosted Tree model on new datasets of demographic data from U.S. counties to generate voting outcome predictions, and to properly consider these predictions when making a decision to push a policy about gambling in a U.S. county.

Business Understanding

The purpose of this report is to build and compare predictive models using a dataset of voting outcome and demographic data to compare the model performances and to identify the best performing model for predicting voting outcome by U.S. county for legalizing gambling. Gambling has historically been a controversial issue from primarily an ethical standpoint, in addition to considerations regarding its impact on society. Therefore, parties that are either in favor of or against the spread of gambling in American culture would be interested in identifying demographic factors that influence the legalization of gambling, to either target or attempt to dissuade specific U.S. counties from voting on gambling ballots.

Identifying a well-performing predictive model for voting outcomes to legalize gambling is necessary to influence U.S. counties either in favor of or against legalizing gambling. An effective predictive model will help identify U.S. counties that may likely pass a measure to legalize gambling, so that parties that are either for or against legalizing gambling can take measures to sway these “high-risk” counties one way or another. Using such a model will allow these parties, such as gambling companies or groups against the legalization of gambling, to deploy campaigns to influence the U.S. counties most susceptible to legalizing gambling to vote either yes or no on the gambling ballots. Additionally, parties such as gambling companies that may want to legalize gambling would want to identify the counties where their effort would likely be successful, before committing to the costs of pushing for legalization.

Data Understanding

The dataset that was used for modeling contains demographic data on 1,287 U.S. counties, with each data point representing a single county. The dataset contains 31 variables, including the voting outcome of each county, which will be used as the target variable for modeling, and demographic variables such as age distribution. The dataset also contains ID variables by state and county. This data is deidentified, and I have full access to the data.

Figure 1 – Statistical Overview of Initial Variables

Name	Type	# Missing values	# Unique values	Minimum	Maximum	25% Quantile	50% Quantile (Medi...	75% Quantile	Mean	Mean Absolute Dev...	Standard Deviation
State No	Number (Integer)	0	18	1	18	8	10	15	10.399	4.229	5.047
County No	Number (Integer)	0	251	1	251	20	41	70	57.272	39.675	54.349
FOR	Number (Integer)	0	1155	44	161,415	812	1,847	4,685	6,460.944	7,452.419	14,766.946
AGAINST	Number (Integer)	0	1181	15	121,925	908	2,017	6,044	7,330.511	8,338.56	15,099.603
TOTAL CASTE	Number (Integer)	0	1237	59	245,523	1,813	3,989	10,771	13,791.455	15,606.578	28,431.805
DEPENDENT VARIABI	Number (Integer)	0	2	0	1	0	0	1	0.426	0.489	0.495
BALLOT TYPE	Number (Integer)	0	2	1	2	1	2	2	1.542	0.496	0.498
POPULATION	Number (Integer)	0	1235	327	1,206,243	6,522	16,126	43,140	52,475.283	58,253.467	109,737.626
PCI	Number (Integer)	0	1177	5,720	44,518	14,203	16,177	18,589	16,652.772	2,860.577	3,827.951
MEDIUM FAMILY INCI	Number (Integer)	0	1216	12,225	55,643	22,904	26,508	31,080	27,398.726	4,982.099	6,403.965
SIZE OF COUNTY	Number (Float)	0	1012	120.8	6,347.8	534	832.5	1,142.2	1,000.488	510.362	778.794
POPULATION DENSIT	Number (Float)	0	1231	0.296	3,404.48	7.009	22.597	65.008	81.573	95.446	221.759
PERCENT WHITE	Number (Float)	0	1075	0.026	0.998	0.7	0.86	0.96	0.804	0.149	0.189
PERCENT BLACK	Number (Float)	0	963	0	0.798	0.001	0.015	0.096	0.081	0.096	0.134
PERCENT OTHER	Number (Float)	0	1071	0	0.973	0.013	0.049	0.14	0.115	0.112	0.167
PERCENT MALE	Number (Float)	0	1107	0.246	0.671	0.483	0.491	0.5	0.494	0.012	0.02
PERCENT FEMALE	Number (Float)	0	1107	0.335	0.754	0.5	0.509	0.517	0.507	0.012	0.02
NO OF CHURCHES	Number (Integer)	0	208	1	738	21	38	70	58.703	42.142	68.467
NO OF CHURCH MEM	String	0	1020	⓪	⓪	⓪	⓪	⓪	⓪	⓪	⓪
PERCENT CHURCH M	String	0	147	⓪	⓪	⓪	⓪	⓪	⓪	⓪	⓪
POVERTY LEVEL	Number (Float)	0	293	3.2	49.9	12.1	16.2	21.5	17.409	5.638	7.214
UNEMPLOYMENT RA	Number (Float)	0	149	0.6	38.5	3.6	5.4	7.6	6.051	2.493	3.41
AGE LESS THAN 18	Number (Integer)	0	1229	87	353,009	1,790	4,568	12,036	14,419.316	15,965.088	30,487.082
AGE24	Number (Integer)	0	1106	16	137,916	417	1,302	4,123	5,204.805	6,099.837	11,700.168
AGE44	Number (Integer)	0	1235	108	437,537	1,712	4,520	12,644	16,497.123	19,097.445	37,260.004
AGE64	Number (Integer)	0	1207	71	202,133	1,328	3,287	8,962	10,298.945	11,286.435	20,948.274
AGE OLDER THAN 65	Number (Integer)	0	1188	21	122,335	1,103	2,401	5,817	6,754.41	7,107.874	13,508.404
MSA	Number (Integer)	0	2	1	2	2	2	2	1.782	0.341	0.413
PERCENT MINORITY	Number (Float)	0	1076	0.002	0.974	0.04	0.14	0.3	0.196	0.149	0.189
NO OF OLDER	Number (Integer)	0	1245	92	324,468	2,428	5,772	14,983	17,053.355	18,348.903	34,050.346
NO OF YOUNGER	Number (Integer)	0	1235	216	875,074	3,424	9,040	25,288	32,994.246	38,194.89	74,520.007

Figure 1 provides general statistics for our initial variables, prior to data preprocessing. The missing values for two of our variables are of note to handle in data preprocessing, as well as the string-type notation for those two variables, when they should be numerical.

Figure 2 – Correlation Matrix of Initial Variables

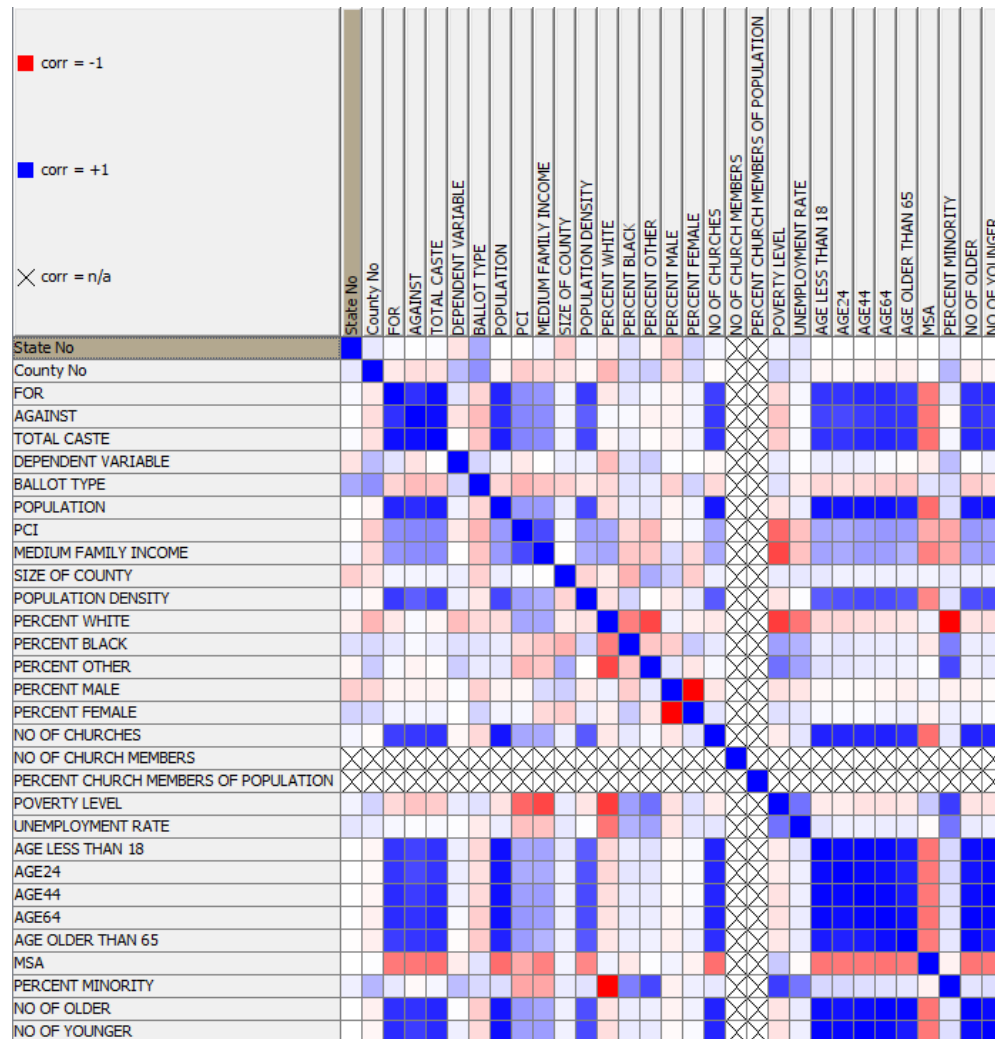


Figure 2 provides a correlation matrix of the majority of the numerical variables, helping to identify variable pairs that have higher correlation either in the positive or negative direction.

Figure 3 – Frequencies of Voting Outcomes

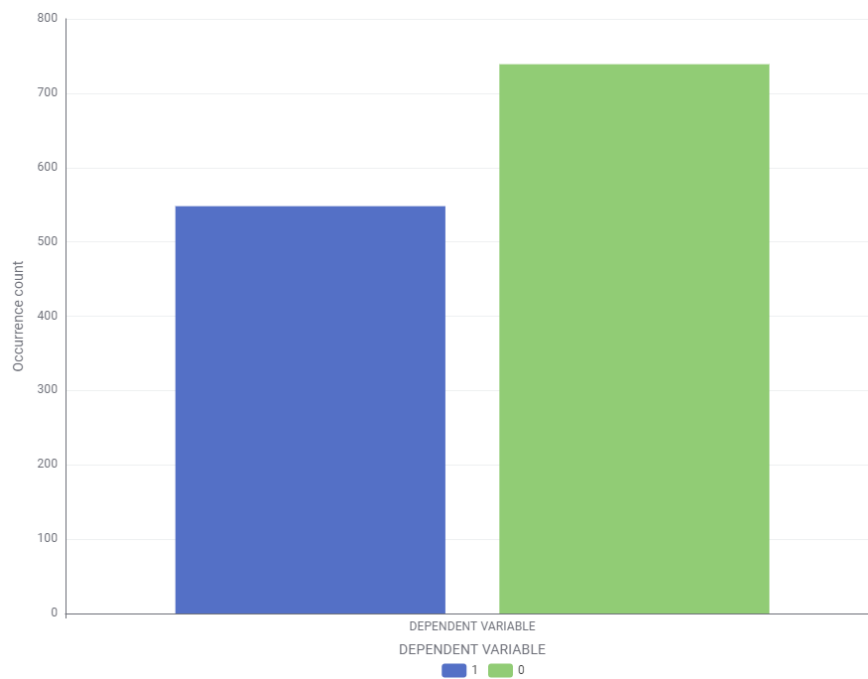


Figure 3 displays the relative frequencies of the 1 (Yes) and 0 (No)) values for whether the county voted to legalize gambling or not. Although the frequencies are relatively similar, I can keep in mind that I should equalize the frequencies in my modeling step.

Data Preparation

Figure 4 – Data Preprocessing

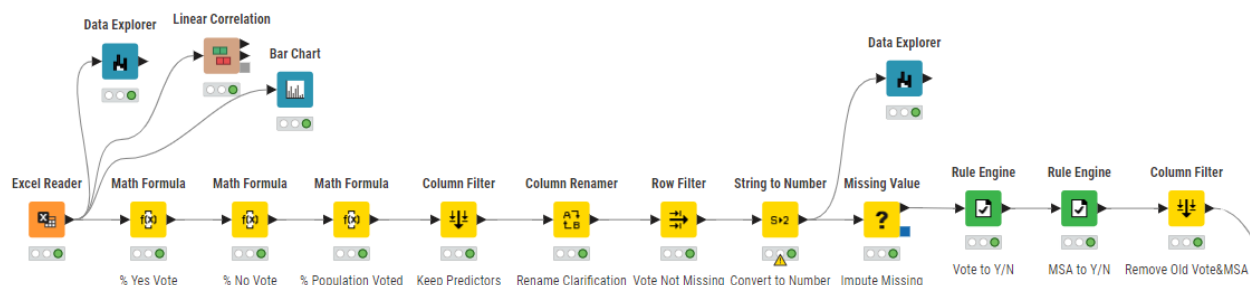


Figure 4 shows the workflow for the data preprocessing steps that were performed. The data, initially contained in an Excel file, is first loaded into KNIME. A series of Math Formula nodes are then used to calculate derived variables: the percentage of the voters that voted Yes, the percentage of the voters that voted No, and the percentage of the population that voted are calculated using the voter population data and the overall population data. A column filter is used to remove variables that are unnecessary to modeling, such as the ID variables. Variables that recorded the number of a group of people when the percentage was already given are considered redundant and filtered out as well. Also, the ballot type variable is removed, since the distinction between gambling and wagering is considered insignificant. The dependent variable column is

renamed to VOTE RESULT for clarification, and the age range variables are named to specify the age range rather than listing a single number, such as AGE24. The rows are filtered to keep just the rows where a voting outcome is recorded, and the two variables that were mistakenly labeled as string variables are relabeled as numerical variables. The one missing value in each of these two variables is imputed with the column's mean value. Lastly, the voting outcome and MSA (Metropolitan Statistical Area) variables are converted to string values by replacing the 1 and 0 values with Yes and No, respectively, and the numerical versions are then deleted. The preprocessed data is then placed into a Color Manager node to color the "Yes" values of the voting outcome variable green and the "No" values red.

Modeling

After being processed, the data is inputted into five different predictive models: Decision Tree, Random Forest, Boosted Tree, Artificial Neural Network, and Logistic Regression. All five models use the data that has been color-coded using the Color Manager node, and all five models are evaluated using Scorer and ROC Curve nodes.

Figure 5 – Decision Tree

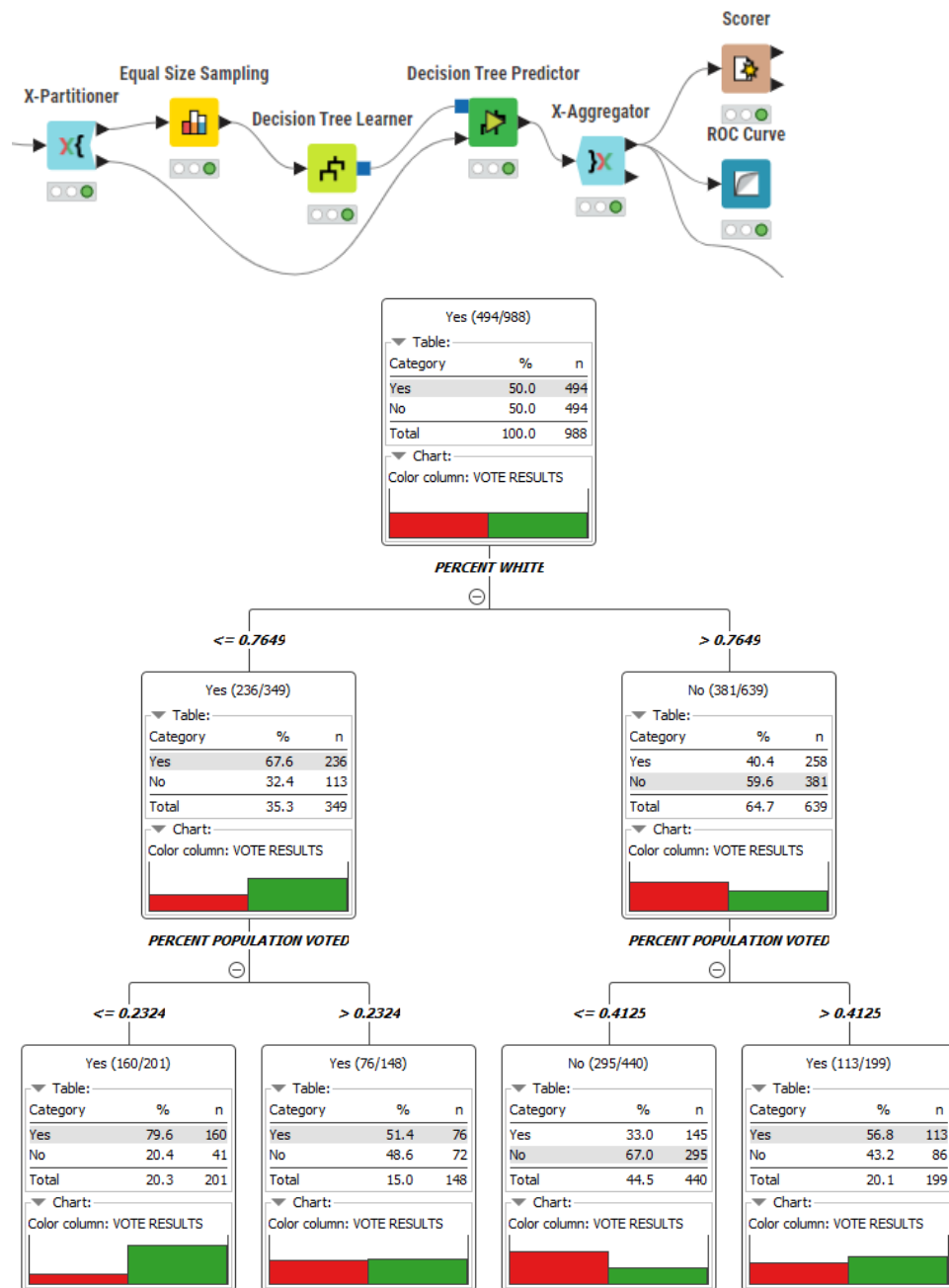


Figure 5 shows the workflow for the decision tree model, as well as the first three levels of the decision tree graphical model. The workflow shows that k-fold cross validation is applied to the model, with 10 folds being created. Equal size sampling is used to equalize the frequencies of voting outcomes for a balanced analysis. The snapshot of the first three levels of the decision tree model shows that the first split occurs where the percentage of white residents is less than or equal to 76.49% or greater than 76.49%. The second set of splits occur based on the percentage

of the county’s population that voted. These splits indicate that the percentage of white residents and the percentage of the population that votes on the gambling ballot are likely the most important variables in predicting the voting outcome for legalizing gambling.

Figure 6 – Decision Tree Performance Metrics

VOTE RES...	Yes	No								
Yes	375	173								
No	234	505								
Correct classified: 880			Wrong classified: 407							
Accuracy: 68.376%			Error: 31.624%							
Cohen's kappa (κ): 0.362%										
RowID	TruePosit... 123 Number (...)	False... 123 Number (...)	TrueNega... 123 Number (...)	FalseNeg... 123 Number (...)	Recall 123 Number (...)	Precision 123 Number (...)	Sensitivity 123 Number (...)	Specificity 123 Number (...)	F-measure 123 Number (...)	Accuracy 123 Number (...)
Yes	375	234	505	173	0.684	0.616	0.684	0.683	0.648	②
No	505	173	375	234	0.683	0.745	0.683	0.684	0.713	②
Overall	②	②	②	②	②	②	②	②	②	0.684

Figure 6 shows the performance metrics of the decision tree model, such as sensitivity, specificity, and accuracy. The sensitivity, specificity, and predictive accuracy values are all around 0.68, indicating that the model has a moderate performance in identifying true positives and true negatives, and in making correct predictions in general.

Figure 7 – Random Forest

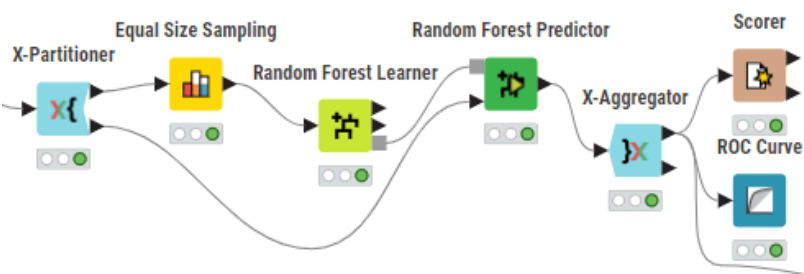


Figure 7 shows the workflow for the random forest model. Like the decision tree model, k-fold cross fold validation is used to split the data into 10 folds for validation, and equal size sampling is applied identically to the decision tree model. The random forest model generated and used the input of 1,000 decision tree models, using the principle of ensemble learning to create a more powerful predictive model.

Figure 8 – Random Forest Performance Metrics

[illegible]

Figure 8 shows the performance metrics of the decision tree model, such as sensitivity, specificity, and accuracy. The sensitivity, specificity, and predictive accuracy values are all around 0.74 to 0.75, indicating that the model has a strong performance in identifying true positives and true negatives, and in making correct predictions in general.

Figure 9 – Boosted Tree

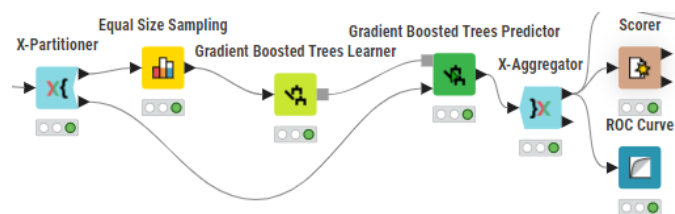


Figure 9 shows the workflow for the boosted model. Like the previous models, k-fold cross fold validation is used to split the data into 10 folds for validation, and equal size sampling is applied identically to the previous models. The boosted tree model is similar to the random forest model in that it builds multiple decision trees; for this model, 100 models are used. The model, however, is unique in that each subsequent tree corrects the errors made by the previous trees, so that the predictive accuracy is sequentially improved.

Figure 10 – Boosted Tree Performance Metrics

[illegible]

Figure 10 shows the performance metrics of the boosted tree model, such as sensitivity, specificity, and accuracy. The sensitivity, specificity, and predictive accuracy values are all around 0.75, indicating that the model has a strong performance in identifying true positives and true negatives, and in making correct predictions in general.

Figure 11 – Artificial Neural Network

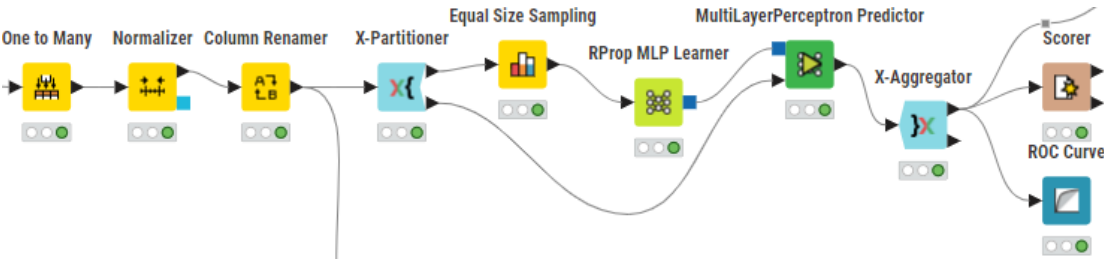


Figure 11 shows the workflow for the artificial neural network model. Unlike the three set-based prediction models prior to this one, the artificial neural network model is a numeric-based model, and thus requires all data values to be numerical. Therefore, one-hot-encoding is used to assign proportional numerical values to non-numerical variables, and min-max normalization is applied to contain the values of every variable between 0 and 1, from minimum to maximum respectively. Like the previous three models, this model uses k-fold cross validation and equal size sampling with the same settings as the prior models. The artificial neural network model is designed to imitate human thinking to perform predictive analysis.

Figure 12 – Artificial Neural Network Performance Metrics

VOTE RES...	Yes	No	
Yes	372	176	
No	223	516	
Correct classified: 888		Wrong classified: 399	
Accuracy: 68.998%		Error: 31.002%	
Cohen's kappa (κ): 0.373%			

RowID	TruePosit... 103 Number (...)	FalsePosi... 103 Number (...)	TrueNega... 103 Number (...)	FalseNeg... 103 Number (...)	Recall 100 Number (...)	Precision 100 Number (...)	Sensitivity 100 Number (...)	Specificity 100 Number (...)	F-measure 100 Number (...)	Accuracy 100 Number
Yes	372	223	516	176	0.679	0.625	0.679	0.698	0.651	🔗
No	516	176	372	223	0.698	0.746	0.698	0.679	0.721	🔗
Overall	🔗	🔗	🔗	🔗	🔗	🔗	🔗	🔗	🔗	0.69

Figure 12 shows the performance metrics of the artificial neural network model, such as sensitivity, specificity, and accuracy. The sensitivity and specificity values range from roughly 0.68 to 0.70, while the predictive accuracy value is similarly at 0.69, indicating that the model

has a moderate performance in identifying true positives and true negatives, and in making correct predictions in general.

Figure 13 – Logistic Regression

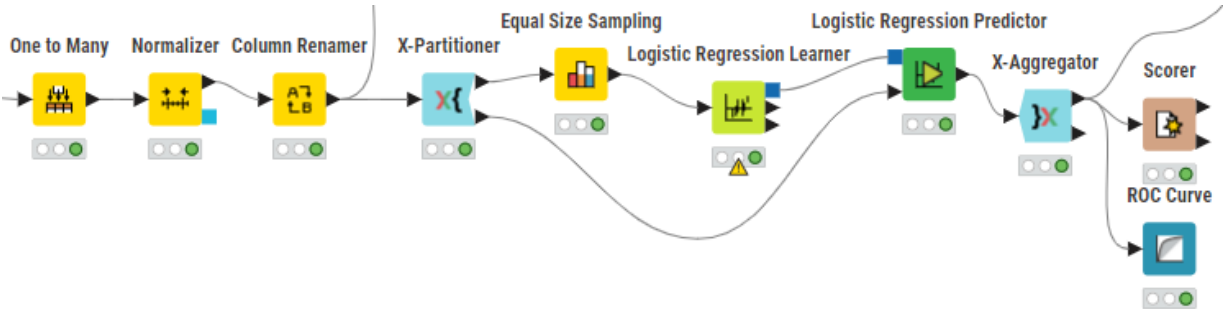


Figure 13 shows the workflow for the logistic regression model. Like the artificial neural network model, this model is a numeric-based model, so one-hot-encoding is used to assign proportional numerical values to non-numerical variables, and min-max normalization is applied to contain the values of every variable between 0 and 1, from minimum to maximum respectively. Like the previous four models, this model uses k-fold cross validation and equal size sampling with the same settings as the prior models. The logistic regression model uses the logarithm of the odds of a county voting Yes on the gambling ballot to estimate the probability of a county voting Yes on a gambling ballot.

Figure 14 – Logistic Regression Performance Metrics

VOTE RES...	Yes	No								
Yes	323	225								
No	208	531								
Correct classified: 854			Wrong classified: 433							
Accuracy: 66.356%			Error: 33.644%							
Cohen's kappa (κ): 0.309%										

RowID	TruePosit...	FalsePosi...	TrueNega...	FalseNeg...	Recall	Precision	Sensitivity	Specificity	F-measure	Accuracy
	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number (...)	Number
Yes	323	208	531	225	0.589	0.608	0.589	0.719	0.599	?
No	531	225	323	208	0.719	0.702	0.719	0.589	0.71	?
Overall	?	?	?	?	?	?	?	?	?	0.664

Figure 14 shows the performance metrics of the logistic regression model, such as sensitivity, specificity, and accuracy. The sensitivity and specificity values range from around 0.59 to 0.72, while the predictive accuracy value is around 0.66, indicating that the model has a moderate performance in identifying true positives and true negatives, and in making correct predictions in general.

Evaluation

Figure 15 – ROC Curves

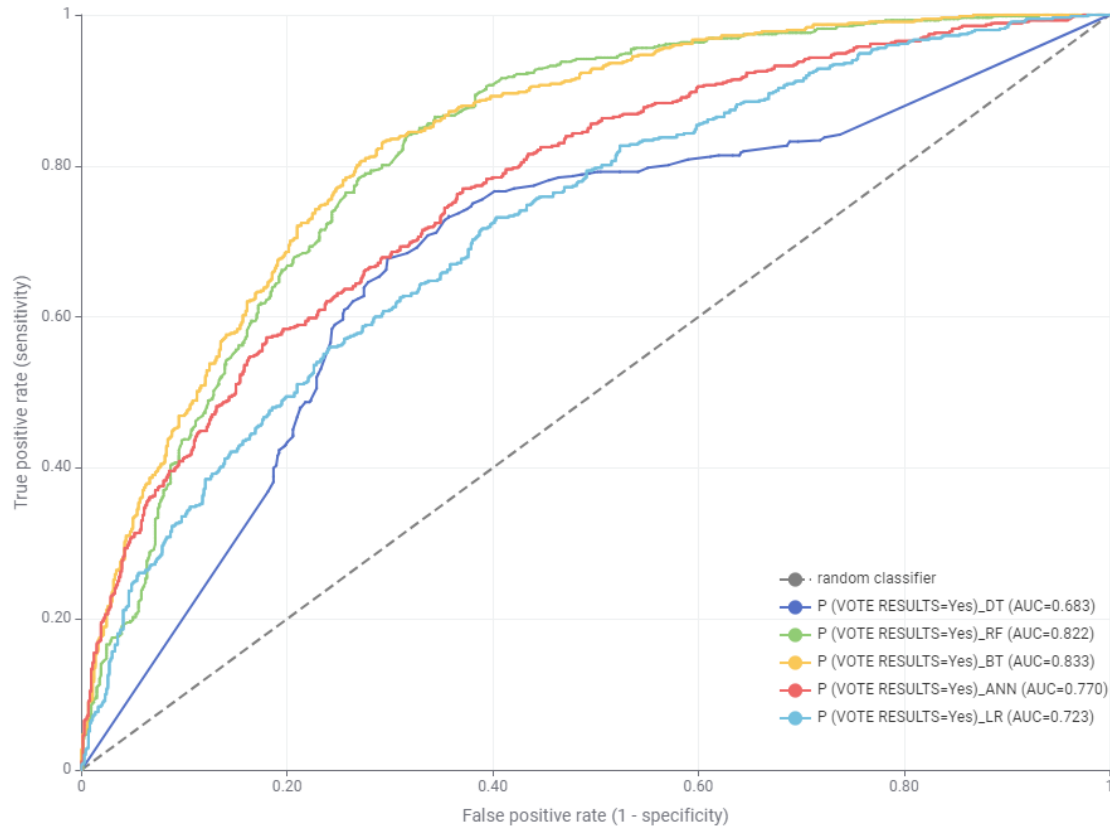


Figure 15 shows the ROC curves for all five models, distinguished by the suffixes in the legend: DT for Decision Tree, RF for Random Forest, BT for Boosted Tree, ANN for Artificial Neural Network, and LR for Logistic Regression. The AUC scores in the bottom right corner show that the boosted tree model yielded the highest AUC value, suggesting that the boosted tree model is the best performing model for distinguishing voting outcomes.

Performance Metric Comparisons:

Based on the performance metrics shown in Figures 6, 8, 10, 12, and 14, the boosted tree model has the highest specificity, sensitivity, and accuracy values, all ranging from roughly 0.75 to 0.77. This indicates that the boosted tree model is the best performing model at identifying true positives and true negatives, and has the best overall predictive accuracy. Due to also having the highest AUC value, the boosted tree model is the model of choice out of the five for predicting voting outcome for legalizing gambling.

Deployment

The model can be deployed for multiple purposes. Firstly, the boosted tree model can be used to predict which U.S. counties are likely to vote to legalize gambling, allowing for gambling companies and other interested parties to target these counties with advertising campaigns. Secondly, the model can be analyzed to assess which variables are the most important predictors of voting outcome; this would allow gambling companies and other interested parties to advertise efficiently, by targeting the demographics most likely to be influenced by the campaign. And finally, the model can be used to identify counties that are unlikely to vote to legalize gambling, and business decisions can be made on whether it would be worth the cost to target certain demographics in those counties with advertisement campaigns to sway the public opinion towards legalizing gambling or not.

Conclusion/Summary of Tools and Techniques

In conclusion, the analysis done in this report highlights the power of predictive modeling, as well as the necessity for careful evaluation to select the best performing model. In this case, one model clearly outperformed the others, but in other cases, some performance metrics may be superior to those of other models while other metrics may be inferior to those of other models. A holistic examination of every relevant metric is a necessity to ultimately decide on which model is best suited for the use case.

KNIME allowed for a variety of tools and techniques to be used for data exploration, data analysis, modeling, and evaluation. The Math Formula nodes allowed for more informative variables to be derived, while redundant and non-informative variables could be filtered out with the Column Filter node. Tools such as the Column Renamer node, the String to Number node, and Missing Value node helped to clean the data, fixing errors such as a wrongly labeled data type and allowing for missing values to be imputed conveniently. Additionally, the Data Explorer node was a powerful tool throughout the data preprocessing step, allowing for the data to be examined at every single step of the preprocessing phase so that errors could be caught easily.

Another useful technique for modeling was k-fold cross validation. Although single-split can provide sufficient predictive accuracy, it works better on larger datasets. In contrast, k-fold cross validation generally yields a higher accuracy on smaller datasets, such as the one used in this report, and tends to produce a higher accuracy in general due to being relatively more

thorough when it comes to splitting up and validating the data. Equal size sampling worked in conjunction with k-fold cross validation by producing a dataset that is balanced on the voting outcomes, so that the analysis would not be skewed towards one of the outcomes.

One-hot encoding and min-max normalization also proved to be helpful for modeling. These techniques allowed for numeric-based modeling to be done on the string-type variables as well, and also for the values to be normalized so that the models would not be skewed in favor of the variables with values of higher magnitude. These two techniques enabled two of the five models and greatly enhanced the quality of the analysis.

Lastly, the Scorer and ROC Curve nodes were effective evaluation tools. The Scorer node produced more than enough performance metrics to be able to compare the model performances, and the ROC Curve provided a nearly definitive metric, the AUC value, for comparing model predictive power. Although some of the tools, such as the Math Formula nodes, ended up not being as useful as they could be due to producing variables that were filtered out, every tool and technique contributed to understanding the data more and conducting a satisfactory analysis for model comparison.

Figure 16 – Complete Workflow

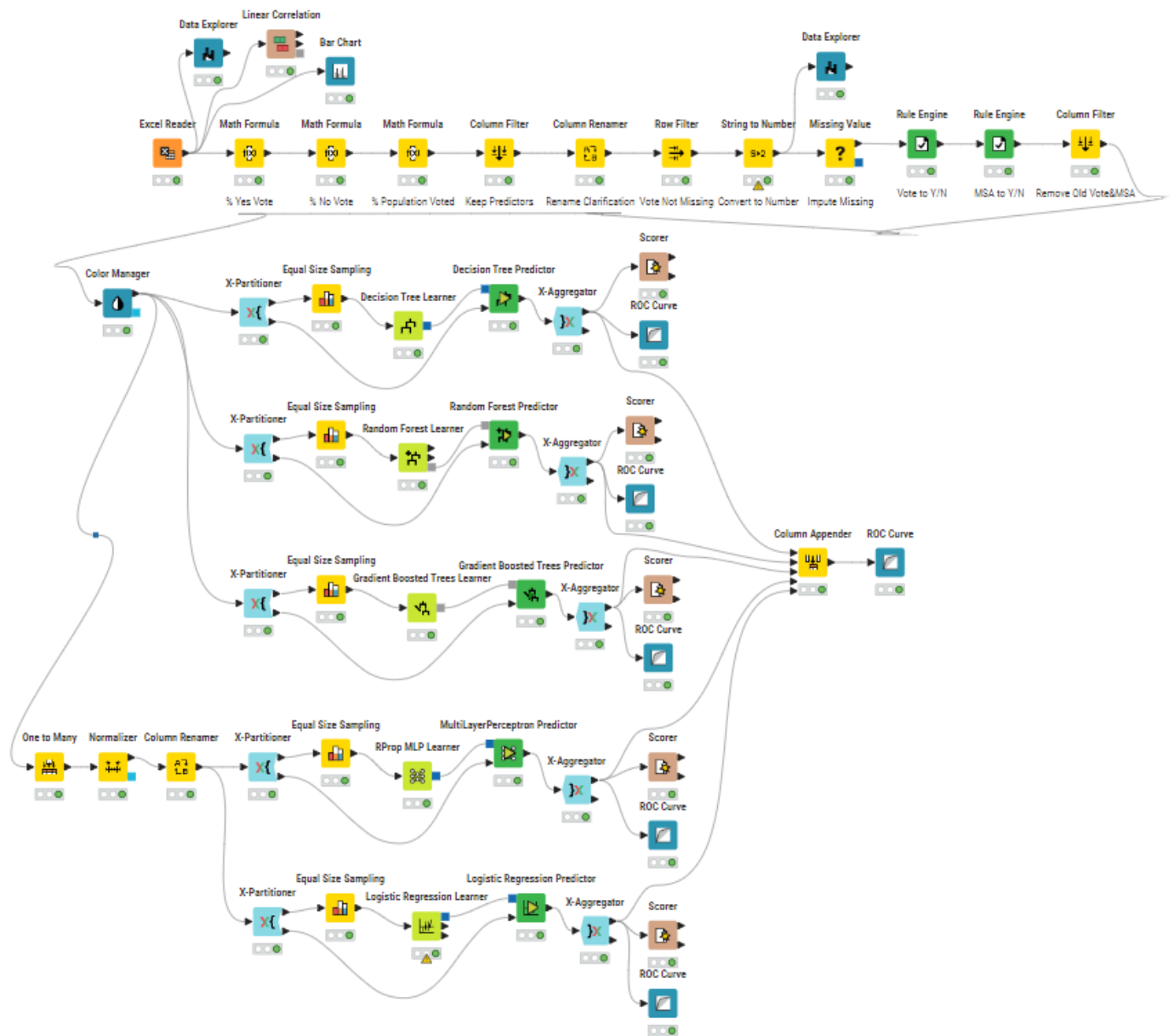


Figure 16 is provided for reference for the Conclusion statement regarding tools and techniques used in this report.